CLASSROOM CONTACT PROGRAMME
(Academic Session : 2025 - 2026)

Achiever COURSE
PHASE : LEADER & ACHIEVER COURSE (SRG)
TARGET : PRE-MEDICAL 2026

Test Type :**MINOR** Test Pattern :**NEET (UG)**

TEST DATE : 07-09-2025

ANSWER KEY

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
A.	3	2	4	3	4	2	3	1	2	2	1	3	2	2	1	1	1	4	4	4	2	1	3	3	1	2	3	4	1	1
Q.	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60
A.	4	4	2	3	4	2	3	1	1	4	1	1	1	1	3	2	1	1	1	3	1	4	1	4	4	3	4	3	4	3
Q.	61	62	63	64	65	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85	86	87	88	89	90
A.	2	2	3	3	4	1	4	2	4	2	2	1	2	1	4	4	2	3	2	1	2	4	4	4	3	4	2	2	4	3
Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120
A.	3	1	4	4	2	2	1	1	3	2	2	3	1	1	3	1	4	2	3	3	3	1	3	4	1	4	2	3	2	3
Q.	121	122	123	124	125	126	127	128	129	130	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	2	3	1	2	3	3	1	1	4	2	1	2	2	2	4	1	4	3	4	1	2	2	3	2	2	1	1	1	1	2
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170	171	172	173	174	175	176	177	178	179	180
A.	1	1	1	3	1	2	4	4	3	3	3	2	3	4	4	2	1	1	2	3	3	2	2	3	3	1	3	2	4	1

HINT – SHEET

1. Ans (3)

Pitch p = $\frac{1\text{mm}}{2}$ = 0.5mm & least count (LC) = $\frac{\text{pitch}}{\text{No. of CSD}} = \frac{0.5\text{mm}}{100}$ = 0.005 mm

In fig (1) No measurement :

There is a Negative zero error

Ze = −(100 -96) × 0.005

In fig (1) During measurement :

MSR = 4 pitch = 4 × 0.5 mm = 2 mm

CSR = 56 CSD = 56 × 0.005 mm = 0.280 mm

Final reading = 2 mm + 0.280 mm −(−0.2 mm) = 2.300 mm

2. Ans (2)

(1) $\left[\frac{dy}{dt}\right] = \frac{L}{T} = LT^{-1}$

(2) $[\alpha^2 t^2] = 1 \quad [\alpha^2] = \frac{1}{T^2} \quad [\alpha] = T^{-1}$

(3) log are dimensionless.

(4) by homogenity principle

$[\alpha[T_2 - T_1]] = 1$

$[\alpha] = \frac{1}{T} = T^{-1}$

3. Ans (4)

Concept :

- Systematic errors remain constant for repeated observations and do not change with the number of trials.
- Random errors vary unpredictably in both sign and magnitude.
- A plot of systematic error vs. number of observations is a horizontal straight line (constant value).

Analysis :

- (a) Correct: Systematic error is constant, so its graph is a horizontal straight line.
- (b) Correct: Systematic errors are either consistently positive or negative.
- (c) Correct: Random errors occur unpredictably and vary in both sign and magnitude.

Final Answer :

All statements (a), (b), and (c) are correct.

Option 4 : (a, b, and c) is correct.

4. Ans (3)

$$PV - Pb + \frac{a}{V} - \frac{ab}{V^2} = RT$$

$$\left[\frac{ab}{V^2}\right] = [PV] = [\text{energy}]$$

5. Ans (4)

$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2}$$

$$\frac{dR}{R^2} = \frac{dR_1}{R_1^2} + \frac{dR_2}{R_2^2}$$

$$\Rightarrow \frac{dR}{R} = R \left\{\frac{dR_1}{R_1} \times \frac{1}{R_1} + \frac{dR_2}{R_2} \times \frac{1}{R_2}\right\}$$

$$\% \text{ error } = 2\left\{\frac{1}{3} + \frac{2}{6}\right\}$$

$$= \frac{4}{3}\% = 1.33\%$$

6. Ans (2)

$g = 4\pi^2 L/T^2$

Here, $T = \frac{t}{n}$ and $\Delta T = \frac{\Delta t}{n}$. Therefore,

$\frac{\Delta T}{T} = \frac{\Delta t}{t}$. the errors in both L and t are the least count errors. Therefore,

$(\Delta g/g) = (\Delta L/L) + 2(\Delta T/T)$

$$= \frac{0.1}{20.0} + 2\left(\frac{1}{90}\right) = 0.027$$

Thus, the percentage error in g is

$100\,(\Delta g/g) = 100\,(\Delta L/L) + 2 \times 100\,(\Delta T/T) = 3\ \%$

7. Ans (3)

Calculation/Explanation:

Given percentage errors:

Substituting

Percentage error in $R = \frac{1}{3}(3\%) + 3 \times (2\%) + 2 \times (1\%)$

$= 1\% + 6\% + 2\% = 9\%$

Final Answer is option (3)

8. Ans (1)

$[Q] = [L^2]$ ....(1)

$$[P] = \left[\frac{dx}{Q - x^2}\right]$$

$$[P] = \left[\frac{L}{L^2}\right] = \left[L^{-1}\right] \text{ ....(2)}$$

$\therefore [PQ] = [L^{-1}L^2] = [L]$

9. Ans (2)

$$\because u = \sqrt{2gH} = \sqrt{32 \times 2g} = \sqrt{64g}$$

$$\because t = \frac{u}{g} = \frac{\sqrt{64g}}{g}$$

$$\therefore 4 = \sqrt{\frac{64}{g}} \quad \therefore g = 4\ m/s^2$$

$$\because S_{n^{th}} = u + \frac{a}{2}(2n-1)$$

$$\therefore S_{4^{th}} = \sqrt{64 \times 4} - \frac{4}{2}(2 \times 4 - 1)$$

$$= 16 - 14 = 2\ m$$

10. Ans (2)

$$\int_0^3 ydx = \text{area under the curve}$$

$$= \frac{1}{2} \times 1 \times 5 + 1 \times 5 + \frac{1}{2} \times 1 \times 5$$

$= 10$

$$\left(\frac{dy}{dx}\right)_{x=0.5} = 5$$

$$\left(\frac{dy}{dx}\right)_{x=2.5} = -5$$

$$\Rightarrow \frac{d^2y}{dx^2} = 0$$

11. Ans (1)

Refer class notes

12. Ans (3)

(P) $v\,(t = 3) = 3^2 - 2(3) + 1 = 4$ m/s

(Q) $a = \frac{dv}{dt} = 2t - 2$

$a(t = 2) = 4 - 2 = 2\ m/s^2$

(R) $v = 0$

$t^2 - 2t + 1 = 0$

$t = 1$ sec

(S) $|\vec{a}_{AV}| = \left|\frac{\Delta\vec{v}}{\Delta t}\right| = \frac{v_f - v_i}{t_f - t_i} = \left|\frac{0-1}{1-0}\right| = 1 m/s^2$

13. Ans (2)

Basic definition based

14. Ans (2)

Position of particle, $x = 3 + 8t - 4t^2$

Velocity of particle, $v = 8 - 8t$

acceleration of particle, $a = -8$

Therefore,

Initial acceleration, $a = -8$ unit

Velocity at the end of 3s,

$v = 8 - 8 \times 3 = -16$unit

distance travelled in 2 seconds.

= Distance in 1st second + Distance in 2nd second

= 4 + 4 = 8 unit

Displacement in 1 second = 4 unit

15. Ans (1)

(Q) ⟶ two values of displacement are not possible at same time

(R) ⟶ distance never decreases with time

(S) ⟶ speed never negative

16. Ans (1)

$V = \frac{dx}{dt} = \frac{a}{b}\left[0 - e^{-bt}(-b)\right]$

$V = ae^{-bt}$

$acc^n = \frac{dV}{dt} = a(-b)e^{-bt}$

at $t = 0$, $V_0 = ae^0 = a$

$a_0 = -abe^0 = -ab$

17. Ans (1)

The distance covered by the ball during the last t seconds of its upward motion = Distance covered by it in first t seconds of its downward motion.

From, $h = ut + \frac{1}{2}gt^2$

$h = \frac{1}{2}gt^2$ [As u = 0 for it downward motion]

18. Ans (4)

Calculation (in brief):

- From the graph: At v = 4 m/s ands = 4 m.

$k = \frac{v^2}{s} = \frac{4}{4} = 1$

then, $v^2 = s$

Differentiating:

$2v\frac{dv}{ds} = 1$ then $v\frac{dv}{ds} = \frac{1}{2}$.

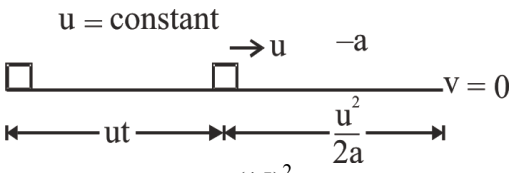
$a = \frac{1}{2}m/s^2$

Final Answer :

option 4 is correct.

19. Ans (4)

Reaction time = 0.4 sec



$\therefore \quad s = (15)(0.4) + \frac{(15)^2}{2 \times 5} = 28.5$ m

20. Ans (4)

$v = \sqrt{49 + x}$, $v^2 = 49 + x$

$2v\frac{dv}{dt} = \frac{dx}{dt} \Rightarrow \quad 2v\frac{dv}{dt} = v$

$\frac{dv}{dt} = \frac{1}{2} = 0.5, \qquad a = 0.5$ m/s^2

21. Ans (2)

Let initial velocity is u and acceleration is a

Applying 3rd equation of motion ⟶

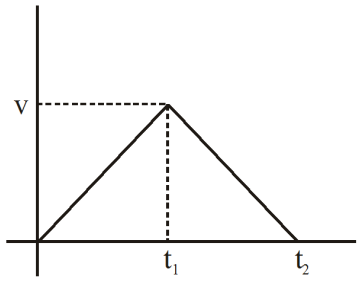
$u^2 + 2a(5) = 6^2$ .....(1)

$u^2 + 2a(5 + 7) = 8^2$ .....(2)

By solving equation (1) and (2)

u = 4 m/s, $a = 2$ m/s^2

22. Ans (1)



$1500 = \frac{1}{2} \times v \times t_1 + \frac{1}{2} \times v \times t_2$

$\frac{3000}{v} = t_1 + t_2$ ... (i)

$s = \frac{v}{t_1}\, t_1 = \frac{v}{5}$

$t_2 = \frac{v}{10}$

$t_1 + t_2 = \frac{3v}{10}$ ... (ii)

from equation (i) and (ii)

$t_1 + t_2 = 30$ sec

23. Ans (3)

Total area

$= \frac{-1}{2} \times 2 \times 10 + \frac{1}{2} \times 2 \times 10 + 1 \times 10 + \frac{1}{2} \times 10 \times 1$

$= -10 + 10 + 10 + 5 = 15$

Average velocity $= \frac{15}{6} = 2.5$ m/s

Average speed for first 4 sec

Area $= + \frac{1}{2} \times 10 \times 2 + \frac{1}{2} \times 2 \times 10 = 20$

$\therefore$ Average speed $= \frac{20}{4} = 5$ m/s

velocity at t = 1 is -5m/s

velocity at t = 4 is 10 m/s

a_{avg} from 1 to 4 sec. is $5m/s^2$

24. Ans (3)

charge in solid sphere - charge of hollow sphere = 0

$$q = \int_0^{R_1} 4\pi r^2 dr \times \frac{\rho_0}{r} = \sigma \times 4\pi R_2^2$$

$$\frac{R_1^2 \rho_0}{2} = \sigma R_2^2$$

$$\therefore \frac{R_2}{R_1} = \sqrt{\frac{\rho_0}{2\sigma}}$$

25. Ans (1)

$$\frac{kq_a z}{(a^2+z^2)^{\frac{3}{2}}} + \frac{kq_b z}{(b^2+z^2)^{\frac{3}{2}}} = 0$$

$$\Rightarrow \left(\frac{b^2+z^2}{a^2+z^2}\right)^{\frac{3}{2}} = \frac{-q_b}{q_a}$$

$$\Rightarrow \frac{q_b}{q_a} = -\left(\frac{4a^2+a^2}{a^2+a^2}\right)^{\frac{3}{2}}$$

$$= -\left(\frac{5}{2}\right)^{\frac{3}{2}}$$

then reciprocal.

26. Ans (2)

From Coulomb's law, the force acting on the particle due to the electrostatic interaction is balanced by the centripetal force required for circular motion:

$$\frac{kQq}{r^2} = \frac{mv^2}{r}$$

Solving for v, we get:

$$v = \sqrt{\frac{kQq}{mr}}$$

Next, using the time period relation $T = \frac{2\pi r}{v}$, we substitute v to get:

$$T = \frac{2\pi r}{\sqrt{\frac{kQq}{mr}}} = 2\pi\sqrt{\frac{mr^3}{kQq}}$$

Squaring both sides:

$$T^2 = \frac{4\pi^2 m r^3}{kQq}$$

Thus, we obtain:

$$r^3 = \frac{kQq}{4\pi^2 m} T^2$$

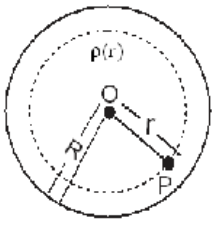
Since $k = \frac{1}{4\pi\epsilon_0}$, substituting k into the equation gives:

$$r^3 = \frac{Qq}{16\pi^3 \epsilon_0 m} T^2$$

Hence, the correct option is Option 2.

27. Ans (3)

By Gauss law



$$\oint \vec{E}.d\vec{s} = \frac{Q_{in}}{\epsilon_o}$$

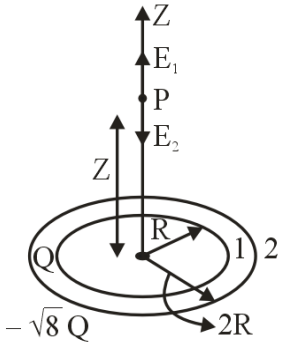
$$E.4\pi r^2 = \frac{\int_0^r \rho_o \left(\frac{3}{4} - \frac{r}{R}\right) 4\pi r^2 dr}{\epsilon_0}$$

$$E4\pi r^2 = \frac{\rho_o 4\pi}{\epsilon_o}\left(\frac{3}{4}\frac{r^3}{3} - \frac{r^4}{4R}\right)$$

$$E r^2 = \frac{\rho_o r^3}{4\epsilon_o}\left\{1 - \frac{r}{R}\right\}$$

$$E = \frac{\rho_o r}{4\epsilon_o}\left\{1 - \frac{r}{R}\right\}$$

28. Ans (4)



Let distance = Z

for $E_R = 0$, $E_1 = E_2$

$$\Rightarrow \frac{KQ(Z)}{(R^2+Z^2)^{\frac{3}{2}}} = \frac{K(\sqrt{8}Q)Z}{(4R^2+Z^2)^{\frac{3}{2}}}$$

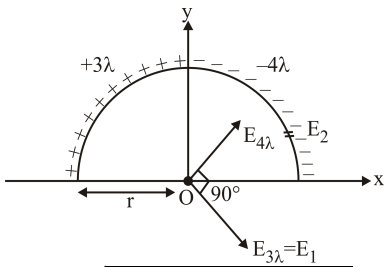
$$\Rightarrow \frac{1}{(R^2+Z^2)} = \frac{2}{(4R^2+Z^2)}$$

$$\Rightarrow 4R^2 + Z^2 = 2R^2 + 2Z^2$$

$$\Rightarrow Z^2 = 2R^2 \Rightarrow Z = \pm\sqrt{2}\,R$$

Distance from origin = $\sqrt{2}\,R$

29. Ans (1)

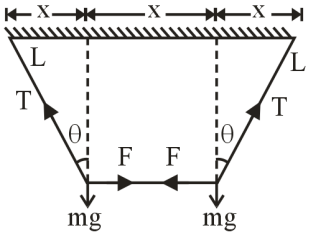


$$E_{net} = \sqrt{E_1^2 + E_2^2 + 2E_1E_2\cos\theta}$$

$E_{net} = 5\sqrt{2}\frac{k\lambda}{r}$ x अक्ष के ऊपर

30. Ans (1)

$\tan\theta = \frac{F_E}{mg}$, $T\sin\theta = F$



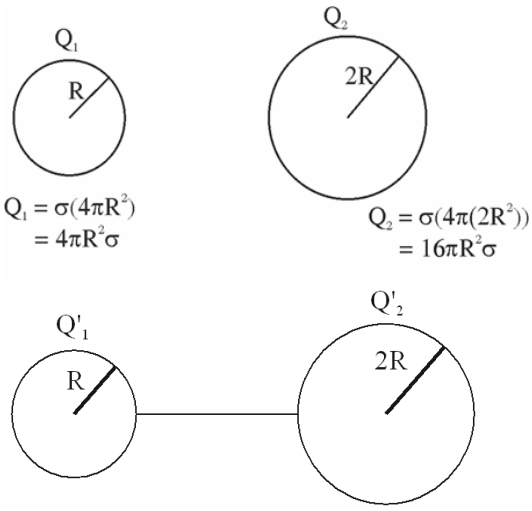
$$\Rightarrow T\left(\frac{x}{L}\right) = \frac{1}{4\pi\varepsilon_0}\frac{Q^2}{x^2}$$

$$T = \frac{1}{4\pi\varepsilon_0}\frac{Q^2L}{x^3}$$

31. Ans (4)

To cover the charge we need 12 surface $\phi = \frac{q}{12\epsilon_0}$

32. Ans (4)



$$\frac{Q'_1}{4\pi\varepsilon_0 R} = \frac{Q'_2}{4\pi\varepsilon_0(2R)}$$

$\therefore\ Q'_2 = 2Q'_1$

$Q'_1 + Q'_2 = Q_1 + Q_2$

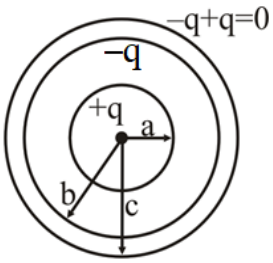
$$\therefore \frac{Q'_2}{2} + Q'_2 = 20\pi R^2\sigma$$

$$\frac{3}{2}Q'_2 = 20\pi R^2\sigma$$

$$\frac{Q'_2}{4\pi(2R)^2} = \frac{2}{3}\cdot\frac{20\pi R^2\sigma}{16\pi R^2}$$

$$\frac{\sigma'}{\sigma} = \frac{5}{6}$$

33. Ans (2)



(i) For r < a $E = \frac{Pr}{3\varepsilon_0} = \frac{Kqr}{a^3}$

(ii) For a < r < b ; E ≠ 0 (Non uniform field)

(iii) For b < r < c ; E = 0 (Inside conductor E = 0 always)

(iv) due to induction

(v) due to induction.

34. Ans (3)

$$\phi_{oneface} = \frac{1}{2}\times\frac{Q}{\varepsilon_0}\times\frac{1}{4} = \frac{Q}{8\varepsilon_0}$$

(half enclosed charge givens flux to four sym. faces

35. Ans (4)

$q_1 + q_2$ ⟵ q_3 ⟶ $F_1 = 2$

$q_2 + q_3$ ⟵ q_1 ⟶ $F_2 = 4$

$q_1 + q_3$ ⟵ q_2 ⟶ $F_3 = 18$

$(q_1 + q_2)q_3 \; \alpha \; +2$ ....(i)

$(q_2 + q_3)q_1 \; \alpha \; -4$ ....(ii)

$(q_3 + q_1)q_2 \; \alpha \; -18$ ....(iii)

on solving (bracket opening)

$\frac{q_1}{q_3} = 2, \frac{q_2}{q_1} = -\frac{3}{4}, \frac{q_2}{q_3} = -\frac{3}{2}$

so ans (4) is correct

36. Ans (2)

Flux linked with sphere = $\vec{E}.d\vec{s}$

since electric field is radial. It is always perpendicular to the surface.

so $\phi = Ar.(4\pi r^2)$

$\phi = A(a)(4\pi r^2)$ (as r = a)

$\phi = A4\pi a^3$

Now according to gauss law

$\phi = \frac{q_{in}}{\epsilon_0} \Rightarrow q_{in} = \phi.\epsilon_0$

so, $q_{in} = A4\pi a^3 \epsilon_0$

37. Ans (3)

$U_i = -\frac{kq^2}{a} - \frac{kq^2}{a} + \frac{kq^2}{2a} = -\frac{3}{2}\frac{kq^2}{a}$

$U_f = +\frac{kq^2}{a} - \frac{kq^2}{a} - \frac{kq^2}{2a} = -\frac{1}{2}\frac{kq^2}{a}$

$\Delta U = U_f - U_i = \frac{kq^2}{a} = \frac{q^2}{4\pi\epsilon_0 a} = W.D$

38. Ans (1)

Net electric field = $3E_1 \cos\alpha$

$= (3)\left(\frac{Q}{4\pi\epsilon_0 a^2}\right)\left(\frac{\sqrt{a^2 - \frac{a^2}{3}}}{a}\right) = \frac{\sqrt{6}\,Q}{4\pi\epsilon_0 a^2}$

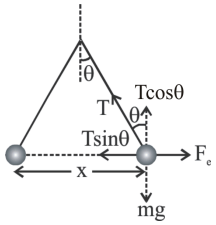
Method -II

E at axis of ring with

$q = 3Q, R = \frac{a}{\sqrt{3}}$

$x = \frac{\sqrt{2}}{\sqrt{3}}a, r = a$

39. Ans (1)



In equilibrium $F_e = T\sin\theta$ ...(i)

$mg = T\cos\theta$ ...(ii)

$\tan\theta = \frac{F_e}{mg} = \frac{q^2}{4\pi\epsilon_0 x^2 \times mg}$

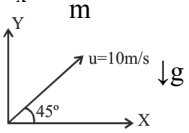
Also $\tan\theta \approx \sin\theta = \frac{x/2}{L}$

Hence $\frac{x}{2L} = \frac{q^2}{4\pi\epsilon_0 x^2 \times mg}$

$\Rightarrow x^3 = \frac{2q^2L}{4\pi\epsilon_0 mg} \Rightarrow x = \left(\frac{q^2L}{2\pi\epsilon_0 mg}\right)^{1/3}$

40. Ans (4)

$a_x = \frac{qE}{m} = \frac{10^{-6} \times 2 \times 10^7}{2} = 10\ m/s^2$



Time of flight $T = \frac{2u\sin\theta}{g} = \frac{2 \times 10 \times \frac{1}{\sqrt{2}}}{10} = \sqrt{2}\,s$

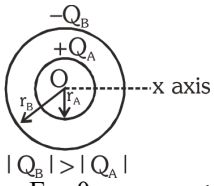
Hence, Range $R = u_x T + \frac{1}{2}a_x T^2$

$R = 10\cos 45° \times T + \frac{1}{2}a_x T^2$

$\frac{10}{\sqrt{2}} \times \sqrt{2} + \frac{1}{2} \times 10 \times 2 = 20m$

(use $g = 10\ m/s^2$)

41. Ans (1)



$|Q_B| > |Q_A|$

$E = 0$; $r < r_A$

$E = \frac{kQ_A}{r^2}$; $r_A < r < r_B$ (along + x)

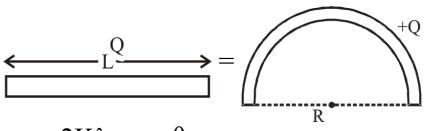
$E = \frac{-k(Q_B - Q_A)}{r^2}$; $r > r_B$ (along – x)

42. Ans (1)

$\because E = \frac{\rho}{3\epsilon_0}r \therefore \frac{\rho}{3\epsilon_0} = \frac{4}{3} \Rightarrow \rho = 4\epsilon_0$

$Q = \rho$ [volume] $= 4\epsilon_0\left[\frac{4}{3}\pi 27\right] = 144\,\pi\epsilon_0$

43. Ans (1)



$$E = \frac{2K\lambda}{R}\sin\frac{\theta}{2}$$

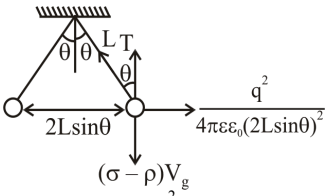
$$E = \frac{2K\lambda}{R} = \frac{2\times 1}{4\pi\epsilon_0}\frac{Q}{L\times\frac{L}{\pi}} \qquad [\pi R = L]$$

$$\boxed{E = \frac{Q}{2\epsilon_0 L^2}}$$

44. Ans (1)

$$\varepsilon_r = \frac{\varepsilon}{\varepsilon_0} \Rightarrow \varepsilon \Rightarrow \varepsilon_r\varepsilon_0 = 81\times 8.85\times 10^{-12}$$

45. Ans (3)



$$\tan\theta = \frac{\frac{q^2}{4\pi\varepsilon\varepsilon_0(2L\sin\theta)^2}}{(\sigma-\rho)V_g}$$

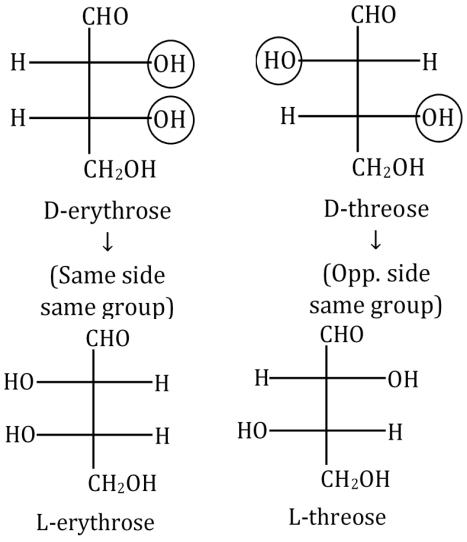
$$= \frac{\frac{q^2}{4\pi\varepsilon_0(2L\sin\theta)^2}}{\sigma V_g}$$

$$\Rightarrow \varepsilon(\sigma-\rho) = \sigma$$

$$\varepsilon = \frac{\sigma}{\sigma-\rho} = \frac{1.6}{1.6-0.8}$$

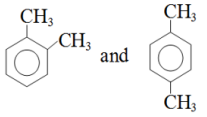
$$\varepsilon = 2$$

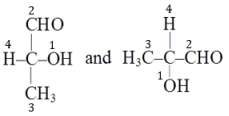
46. Ans (2)

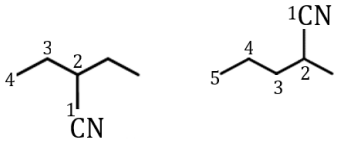


47. Ans (1)

(A) $CH_3-\overset{O}{\overset{\|}{C}}-O-CH_2-CH_2-CH_3$ and $CH_3-CH_2-\overset{O}{\overset{\|}{C}}-O-CH_2-CH_3$ Metamers

(B)  Position isomers

(C)  Enantiomers

(D)  Chain Isomers

48. Ans (1)

49. Ans (1)

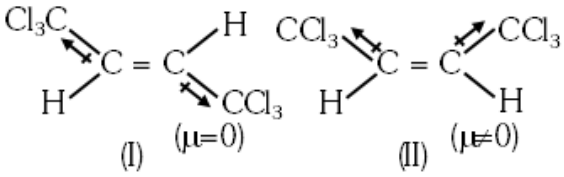
50. Ans (3)

Due to intramolecular H-bond (A) is more stable than B

51. Ans (1)

Fact

52. Ans (4)



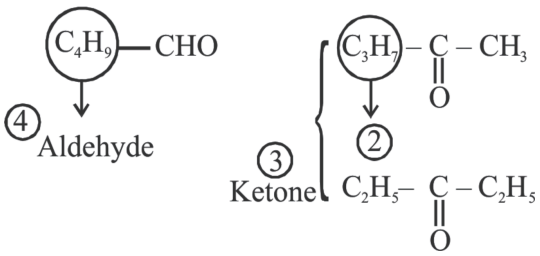
Dipole moment : I > I (Additive moment vector in II)

Boiling moment : II > I (II is more polar)

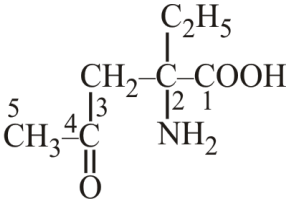
Melting point : I > II (I is symmetrical, more packing)

Stability : I > II (In I, less steric repulsion)

53. Ans (1)



55. Ans (4)



2-Amino-2-ethyl-4-oxo pentanoic acid

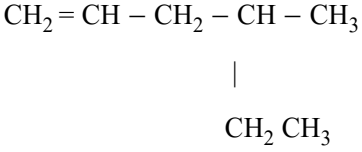
(Substituents arranged in alphabetical order)

56. Ans (3)

Question is Asking About :-

The question asks correct IUPAC name of given compound.

Given Data :-



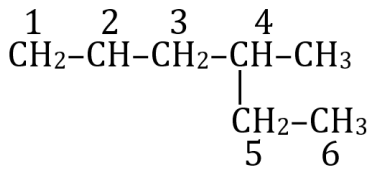
Concept :- IUPAC Nomenclature

Identify the PCC containing multiple bond.

Number the carbon chain that multiple bond gets lowest possible number.

Identify the substituents.

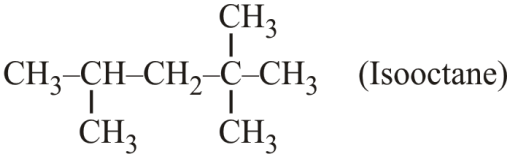
Solution/Explanation/Calculation :-



IUPAC Name is :- 4 – Methyl hex –1–ene

Hence, option (3) is correct.

58. Ans (3)



60. Ans (3)

61. Ans (2)

$\Delta x.\Delta P \geq \frac{h}{4\pi}$

OR

For calculation use

$\Delta x.\Delta P = \frac{h}{4\pi}$

Given

$\Delta P = 3 \times \Delta x$

$\Delta x = \frac{\Delta P}{3}$

$\frac{\Delta P}{3}.\Delta P = \frac{h}{4\pi}$

$(\Delta P)^2 = \frac{3h}{4\pi}$

$(m\Delta V)^2 = \frac{3h}{4\pi}$

$(\Delta V)^2 = \frac{3h}{4\pi m^2}$

$\Delta V = \frac{1}{2m}\sqrt{\frac{3h}{\pi}}$

62. Ans (2)

For porbital $\ell = 1$

Orbital angular momentum

$\sqrt{\ell(\ell+1)}\,\frac{h}{2\pi}$

$\sqrt{1(1+1)}\,\frac{h}{2\pi} = \frac{\sqrt{2}h}{2\pi}$

63. Ans (3)

$n_1 + n_2 = 4$

$n_2 - n_1 = 2$

$n_1 = 1 ; n_2 = 3$

$\bar{\nu} = \frac{1}{\lambda} = R_H \times z^2\left[\frac{1}{n_1^2} - \frac{1}{n_2^2}\right]$

$\bar{\nu} = \frac{1}{\lambda} = R_H \times (3)^2\left[\frac{1}{1^2} - \frac{1}{3^2}\right] = 8R_H$

64. Ans (3)

NCERT Pg. # 55

65. Ans (4)

For maximum wavelength energy gap should be minimum.

$\Delta E = \frac{hc}{\lambda}$

68. Ans (2)

For 8.8g of CO_2

no. of molecule = $\frac{8.8}{44} \times 6.022 \times 10^{23}$

$= 12.044 \times 10^{22}$

For 5.6 L of CO_2 at STP

mole = $\frac{5.6}{22.4} = 0.25$

no. of O-atom = $\frac{96}{32} \times 2 \times 6.022 \times 10^{23}$

$= 36.138 \times 10^{23}$

For 49 g of H_2SO_4

mole = $\frac{49}{98} = 0.5$

69. Ans (4)

NCERT - XI, Pg.# 28 (part 1) 2022

70. Ans (2)

2×35.5 g Cl combine = 32 g S

$\therefore$ 1 g Cl combine = $\frac{32}{2 \times 35.5}$

$\therefore$ 35.5 g Cl combine = $\frac{32}{2 \times 35.5} \times 35.5$

71. Ans (2)

$\frac{49.65}{100} = \frac{142}{142 + 18x}$

$x = 8$

72. Ans (1)

$CaCO_3 \longrightarrow CaO + CO_2$

$\downarrow$

100g $\leftarrow$ 44g

$\frac{100}{44} \leftarrow$ 1g

$\frac{100}{44} \times 17.6 \leftarrow$ 17.6g

% purity = $\frac{\frac{100 \times 17.6}{44}}{50} \times 100 = 80\%$

73. Ans (2)

C_3O_2 : $3x - 4 = 0 \Rightarrow x = \frac{4}{3}$

74. Ans (1)

The balanced redox reaction is

$2MnO_4^- + 5C_2O_4^{2-}$

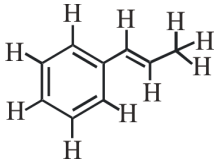
$+16H^+ \longrightarrow 2Mn^{2+} + 10CO_2 + 8H_2O$

75. Ans (4)

76. Ans (4)

NCERT (XIth) (2002) Pg. # 153

77. Ans (2)

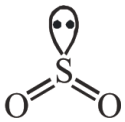


19σ and 4π

78. Ans (3)

SF_4 4σ +1lp = 5 sp^3d

79. Ans (2)



sp^2

1 = pπ – pπ

1 = pπ – dπ

80. Ans (1)

Concept

81. Ans (2)

SiF_6^{-2} due to crowding.

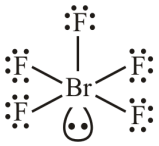
82. Ans (4)

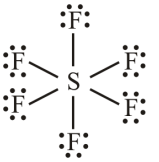
XeF_2

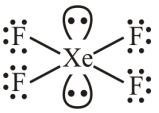
sp^3d

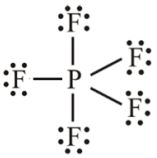
Linear

83. Ans (4)

(A)  sp^3d^2

(B)  sp^3d^2

(C)  sp^3d^2

(D)  sp^3d

84. Ans (4)

$NH_3 + H^+ \longrightarrow NH_4^+$

sp^3 sp^3

(1LP) (OLP)

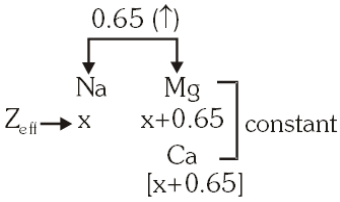
85. Ans (3)

Ge > Br > C > F

86. Ans (4)

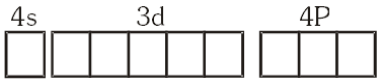
Li^+ & Mg^{2+} [due to diagonal relationship]

87. Ans (2)



88. Ans (2)

IV period :



no. of orbital = 9

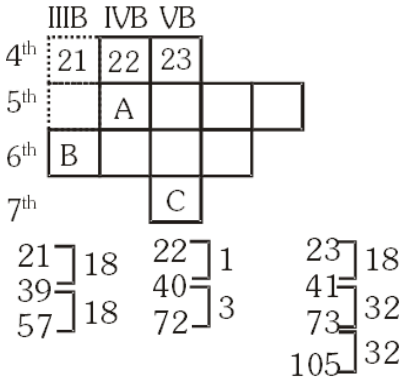
(one orbital occupy 3 electron)

then no. of elements = 9 × 3 = 27

89. Ans (4)

3d transition series = 10 elements (21 to 30)

90. Ans (3)



93. Ans (4)

NCERT-XI, Pg. # 33,32,30,31

94. Ans (4)

Spirogyra, Albugo, Penicillium, Volvox, Mucor, Ustilago

95. Ans (2)

NCERT Pg # 19, 32, 33

97. Ans (1)

99. Ans (3)

103. Ans (1)

Solution:

The Correct answer is : 1. Cell wall

104. Ans (1)

106. Ans (1)

107. Ans (4)

NCERT XI pg.# 21

108. Ans (2)

NCERT XI Pg. # 23

(2.3)

109. Ans (3)

NCERT XI Pg. # 26,27

(2.6)

110. Ans (3)

NCERT XI Pg. # 23

(2.3)

Ascospore, basidiospore and oospores

111. Ans (3)

NCERT XI Pg. 23, (2.3.1)

112. Ans (1)

NCERT Pg. # 18

113. Ans (3)

NCERT Pg. # 14

114. Ans (4)

NCERT pg. #

115. Ans (1)

116. Ans (4)

125. Ans (3)

NCERT Page No. 131

132. Ans (2)
NCERT XI Pg. # 4

133. Ans (2)
NCERT XII Pg # 4

136. Ans (1)
NCERT (XI) Pg. # 272

137. Ans (4)
NCERT XI, Pg # 189

138. Ans (3)
NCERT-XIth Pg # 273 [E]

139. Ans (4)
NCERT-XI Pg. # 272

140. Ans (1)
At time of birth child was alive because of expansion of lung ⟶ It's piece kept floating.

141. Ans (2)
NCERT (XI) Pg # 274

142. Ans (2)

143. Ans (3)

144. Ans (2)
NCERT (XI) Pg. # 273

145. Ans (2)
NCERT (XIth) Pg. # 273, Para-1

146. Ans (1)
NCERT-XII, Pg. # 186

147. Ans (1)
NCERT Pg. # 273

148. Ans (1)
NCERT-XI, Page No. 190

149. Ans (1)
NCERT-XI, Pg # 271

150. Ans (2)

151. Ans (1)

152. Ans (1)

153. Ans (1)
NCERT Page 148,149

154. Ans (3)
NCERT XI, Pg # 150

155. Ans (1)
Question Explanation : Correct statement
Concept : This question is based on Lipids
Solution :

- Lipids- water insoluble
  - R-Group $-CH_3$, $-CH_2$, $-C_2H_5$
- Palmitic Acid - 16C
  - A, B, C are correct

Final Answer : option (1)

156. Ans (2)
NCERT-XI, Pg. # 144

157. Ans (4)
Solution/Explanation:
Analyzing the Options:

1. Maltose: Maltose is made up of two glucose molecules.
2. Lactose: Lactose is made up of glucose and galactose.
3. Sucrose: Sucrose is made up of glucose and fructose.

The correct answer is option 4. ALL OF THE ABOVE.

158. Ans (4)
NCERT XI, Pg # 145

159. Ans (3)
NCERT XI, Pg # 148 Polysaccharide paragraph

160. Ans (3)
NCERT (XI) Pg. # 102, 103

161. Ans (3)
NCERT XI Pg. 112

162. **Ans (2)**
NCERT XI, Pg. No. 103

163. **Ans (3)**
NCERT Pg # 103

164. **Ans (4)**

165. **Ans (4)**

166. **Ans (2)**

167. **Ans (1)**
NCERT Pg.#119

168. **Ans (1)**
NCERT Pg # 119

169. **Ans (2)**
NCERT XI Pg # 80

170. **Ans (3)**
NCERT Pg.# 82

171. **Ans (3)**
Both nucleus and haemoglobin

172. **Ans (2)**
Ventral side of skin is generally olive green

173. **Ans (2)**
2500-3000

174. **Ans (3)**
a, b, c

175. **Ans (3)**
NCERT Pg.# 114

176. **Ans (1)**
OLD NCERT XI, Pg. No. 113

177. **Ans (3)**
a, f, c, b, d, e

178. **Ans (2)**
NCERT Pg. # 115

179. **Ans (4)**
NCERT Page # 111

180. **Ans (1)**

✂ ---- ------✂ ---------- ✂ ----------✂ ----------✂--- ----------✂--- ----------✂---- ----

Note : This is **CORRECTION SLIP** for any doubtful error in paper, which student can submit to class counter person on next day of test , by tearing as per marking done here.

CORRECTION SLIP				
Batch Name	**Test Type**	**Test Date**	**Paper code**	**Student Form No.**
Question No.	**Subject**	**Given Answer**	**Correct Answer**	**Reason/ Remark**